REAL-DATA PILOT

Pancreatic cancer: from concept to real data

A bounded, auditable technical pilot using NCI GDC / TCGA-PAAD pathology and RNA-seq

Real images.
Traceable labels.
Explicit evidence limits.

A research step toward a future multimodal clinical platform

24

real patients processed

48 verified source files
768 retained tissue tiles
3.55 GB of source data

No validated clinical stage, EMT progression or treatment-response prediction is claimed.

The real-data pilot is separate from the synthetic demo

Demonstration metrics are not reused in this report.

01

Real source data

Actual diagnostic SVS slides and STAR-Counts gene-expression files downloaded from GDC.

Every raw file passed size and MD5 verification.

02

Pilot-grade controls

Configuration and cohort plan saved before analysis.

Patient/site traceability, physical-resolution metadata, rejection logs and a fixed baseline comparison.

03

Honest endpoints

Primary aim: acquisition, linkage and tiling feasibility.

Secondary aim: exploratory bulk RNA association. Stage and treatment claims remain outside the evidence.

TCGA-PAAD: explicit selection rules

The project name alone is insufficient to establish histological eligibility.

01

Clinical metadata

Unique primary pancreatic diagnosis; morphology 8500/3; prior treatment recorded as No.

Unrelated metastases and non-ductal histologies are not used as the primary label.

02

Image + RNA records

Primary diagnostic 01Z/DX slide and primary-tumor STAR-Counts RNA file.

Join by patient. Exact tissue-block and spatial correspondence are not established.

03

Bounded sampling

Smallest eligible slide per patient, then first 24 by size; 4-GB cap.

This convenience sample favors small images and certain source sites. It is not representative.

Sources: GDC API [1,2]; NCI TCGA acknowledgement [3]. References on the final slide.

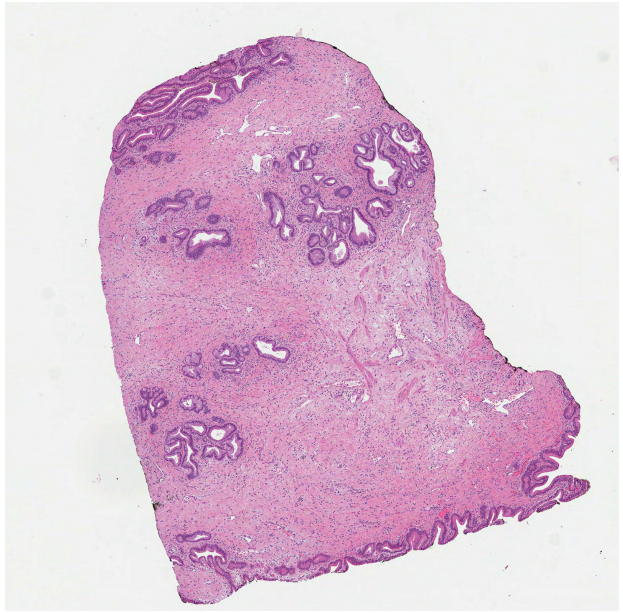
Track acquisition and processing denominators

The full inventory and sequential exclusions are preserved in the acquisition plan.

Area	Measured result	Meaning
Source inventory	649 queried files; 466 slide files.	TCGA-PAAD slide-image and gene-expression file metadata.
Eligible population	142 unique metadata-eligible patients.	One eligible diagnostic slide and primary-tumor RNA record per patient.
Pilot selection	24 patients; 48 verified files.	Selection by size for technical feasibility, not by stage or outcome.
Processing	24 evaluable patients; 0 processing exclusions.	768 tiles retained. Expert tumor review remains outstanding.

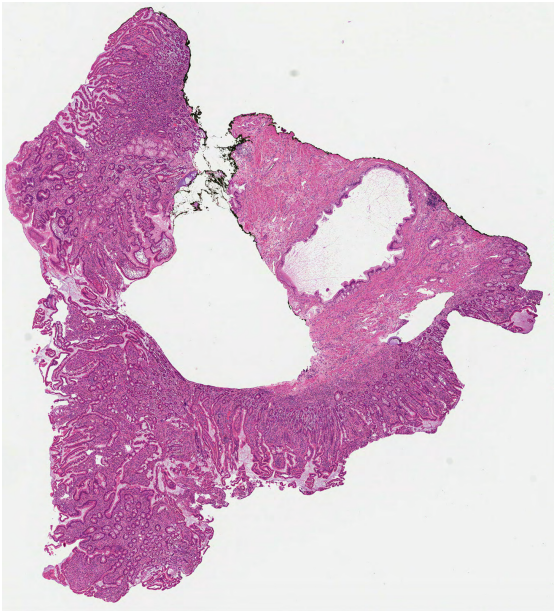
Actual pancreatic diagnostic slide overviews

Overview images demonstrate real data acquisition; they are not tumor annotations.



TCGA-HZ-7926

Native MPP 0.2520; unreviewed tissue



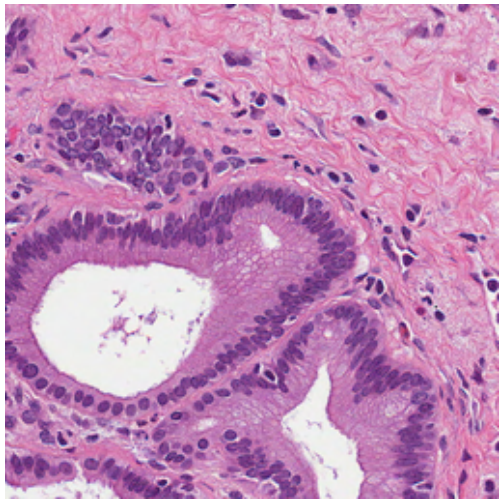
TCGA-HZ-7924

Native MPP 0.2520; unreviewed tissue

Image source: TCGA Research Network, accessed through NCI GDC. Patient/file UUIDs are in the manifest.

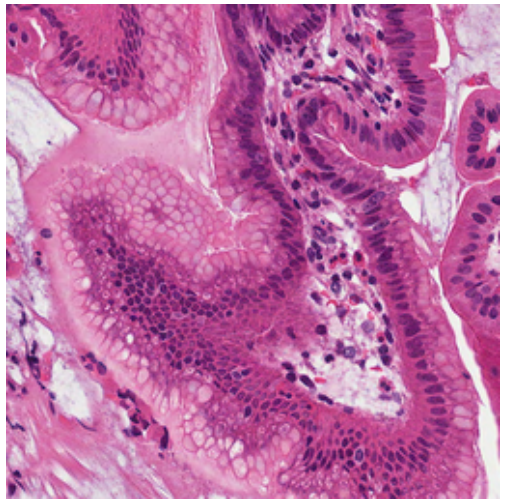
Tissue tiles at a documented physical scale

Nominal 1 micrometer per pixel; 256 x 256 pixels. Tumor content has not been expert-adjudicated.



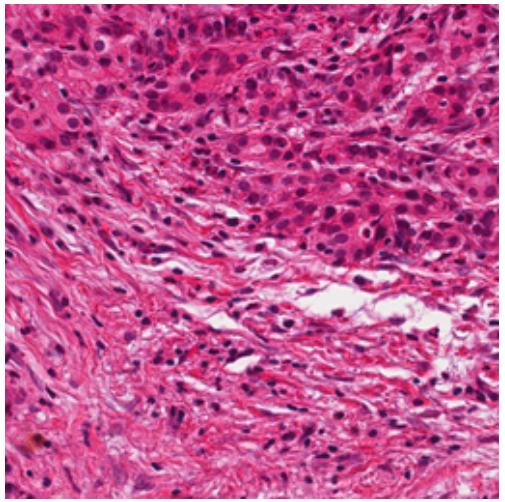
TCGA-HZ-7926

x=5080, y=10160
Tissue fraction 90%



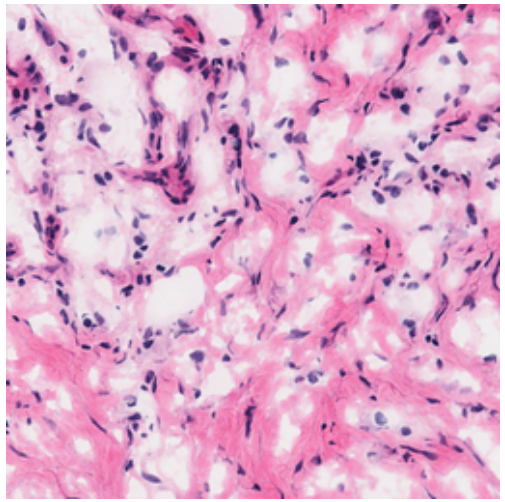
TCGA-HZ-7924

x=2032, y=11176
Tissue fraction 95%



TCGA-HV-A5A5

x=15210, y=11154
Tissue fraction 98%



TCGA-FB-AAPS

x=28392, y=10140
Tissue fraction 81%

Representative acquisition examples from the first four selected patients; no EMT or malignant-cell annotation is implied.

The stage distribution does not support four-stage training

Stage values are database reference labels, not model predictions.

Broad reference stage	Observed count / decision	Pilot implication
Stage I	2 patients	Too few for a credible stage-I performance estimate.
Stage II	21 patients	Dominant class; majority-class predictions can look deceptively accurate.
Stages III / IV	1 / 0 patients	Rare or absent advanced stages prevent four-stage evaluation.
Model decision	NOT_TRAINED	Exact labels and staging editions retained: {'7th': 24}. No retrospective relabeling to balance classes.

A prespecified exploratory bulk RNA composite

This endpoint is neither a validated EMT assay nor a percentage of cells undergoing EMT.

01

Mesenchymal panel

VIM, FN1, CDH2, ZEB1, ZEB2, SNAI1, SNAI2 and TWIST1.

Compute the mean of log2(TPM + 1) across these eight genes.

02

Epithelial panel

CDH1, EPCAM, KRT8, KRT18 and KRT19.

Subtract their mean log2(TPM + 1) from the mesenchymal mean. All 13 genes must be present.

03

Interpretation limits

Unbounded expression contrast, not a universal biological scale.

Bulk signal can reflect fibroblasts. Different tissue blocks can weaken image/RNA correspondence. No progression measurement.

Conceptual EMT limitations follow the TEMTIA consensus [4]. This specific composite is exploratory.

Hold out contributing sites, not individual tiles

Internal site-grouped evaluation; not independent external clinical validation.

01

One row per patient

Average 18 fixed RGB summary features over retained tissue tiles.

No stage or RNA target enters image selection. No cross-patient duplicate tile hashes were detected.

02

Fixed analysis

StandardScaler + Ridge(alpha=10), fitted within each training fold.

Leave one TCGA source-site code out at a time. Compare against the same training-fold mean.

03

Limited uncertainty

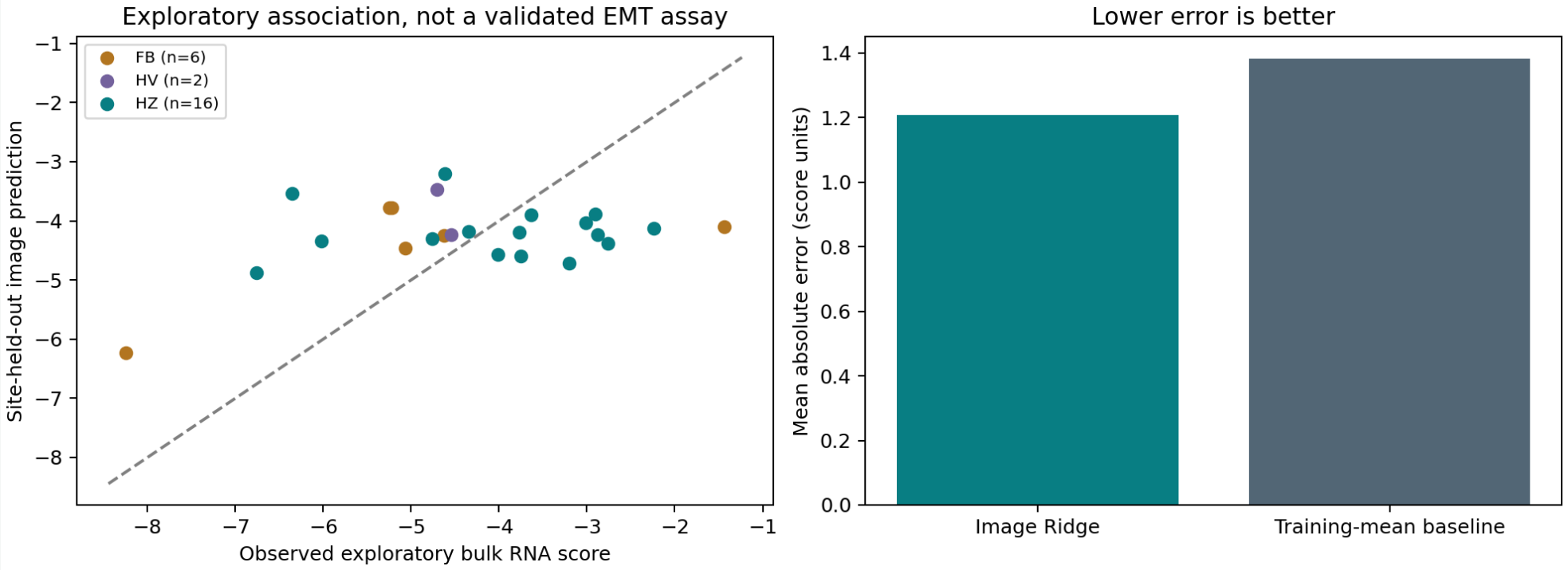
Source-site groups: FB: 6, HV: 2, HZ: 16.

Patient bootstrap intervals use fixed out-of-fold predictions. They omit model-refit and site-cluster uncertainty.

Source-site codes identify tissue contributors; they do not necessarily identify scanning laboratories.

Exploratory results from real data

Report the baseline comparison regardless of direction; do not tune after seeing the result.



The descriptive interval includes no improvement; predictive advantage is not established.

Quantify error without claiming clinical performance

All values below refer to an exploratory expression contrast, not stage accuracy.

Area	Measured result	Meaning
Image prediction	MAE 1.208; RMSE 1.409.	Descriptive MAE interval: 0.937 to 1.496 score units.
Mean baseline	MAE 1.383.	Training-only mean fitted independently in every site-held-out fold.
Paired improvement	Baseline MAE minus model MAE: 0.176.	Descriptive interval: -0.135 to 0.476. Includes zero; an advantage is not established.
Association	R-squared 0.126; Spearman r 0.083.	Small sample; no clinical threshold, biological assay validation or causal interpretation.

Technical completion is distinct from clinical validity

Acceptance rules were specified in the pilot protocol before analysis.

Area	Measured result	Meaning
Raw-file integrity	PASS	48 source files checked against repository size and MD5.
Evaluable patients	PASS	24/24 processed; required at least 22. Resource-based feasibility target.
Image traceability	PASS	At least eight tiles per patient; coordinates, MPP and hashes retained.
Clinical readiness	NOT ESTABLISHED	No expert tumor adjudication, valid four-stage cohort, prospective evaluation or treatment-response evidence.

Use transparent reporting, not a compliance label

Reporting crosswalk is informed by TRIPOD+AI; no certification or full compliance is asserted.

01

Prespecified + auditable

Versioned configuration, selection plan and fixed endpoints.

Inventory, download receipts, code hash, sample linkage, exclusions and analysis files remain reviewable.

02

Data separation

Patient aggregation and source-site folds prevent direct tile leakage.

Training-only scaling and mean baseline. No feature or hyperparameter tuning on held-out predictions.

03

Open limitations

Convenience sampling and tiny site groups disclosed.

Expert review, local governance, patient involvement, external validation and clinical utility remain outstanding.

Reporting reference: TRIPOD+AI, BMJ 2024;385:e078378 [5].

The pilot establishes a data pipeline, not a clinical platform

Avoid promoting a feasible acquisition workflow into unsupported medical claims.

01

Anatomical stage

A primary-tumor image lacks the complete metastatic and vascular assessment.

This cohort cannot estimate performance across all four stages. No stage classifier was trained.

02

EMT progression

The bulk RNA composite may reflect stromal content and spatial mismatch.

Longitudinal matched assays and malignant-cell localization are needed to study progression.

03

Therapeutic benefit

Treatment exposure, outcomes and a suitable causal or prospective study are absent.

The earlier therapeutic panel remains educational, based on clinician-confirmed status.

Spend the next phase on labels and cohort quality

Recommended next work; not completed or funded by this technical pilot.

Work package	Next deliverable	Decision purpose
Pathology review	Expert-check a pilot subset, then all retained tumor regions.	Record tumor/stroma fractions, artifacts, adequacy and interobserver agreement.
Biological validation	Confirm malignant-cell EMT with spatial or multiplex evidence.	Audit tissue-block/timepoint match; assess assay reliability and stromal confounding.
Model comparison	Expand with prespecified sample size and external center(s).	Compare clinical-only, stain/site-only and pathology-encoder baselines under locked evaluation.
Decision gate	Proceed only if the new endpoint and model add reliable value.	Negative or uncertain results trigger endpoint/model revision, not stronger clinical language.

Advance only when each evidence gate is met

Illustrative planning horizons; no approval or deployment date is promised.

Illustrative horizon	Evidence stage	Required next capability
0-6 months	Technical pilot + expert label review	Use this pipeline to establish cohort access, annotation quality and biological validity.
6-18 months	Expanded retrospective / external study	Power the intended endpoint; lock the model; quantify subgroup errors and calibration.
18-30 months	Prospective silent evaluation	Test hospital workflow, review time, mismatch handling, uncertainty and operational failure.
30+ months	Defined-use clinical translation	Appropriate clinical-impact evidence, jurisdiction-specific review, security and lifecycle monitoring.

Every displayed result has a local evidence trail

Raw data remain outside the source-code archive; reproducible download scripts are included.

Purpose	Files	Evidence retained
Acquire	acquire_pilot.py; pilot_config.json	GDC inventory, fixed cohort plan and checksum-verified source downloads.
Analyze	run_real_pilot.py; requirements-pilot.txt	SVS tiling, gene composite, site-grouped evaluation and environment snapshot.
Review	pilot_app.py; outputs/real_pilot/	Real images, manifests, gene TPMs, OOF predictions, figures and pilot_results.json.
Audit	docs/PILOT_PROTOCOL.md; reporting checklist	Scope, rules, acceptance gates, missing work and attribution. Local plan/code hashes in results.

Data and reporting references

Sources checked 13 November 2026. No TCGA endorsement is implied.

[1] GDC API search and retrieval

https://docs.gdc.cancer.gov/API/Users_Guide/Search_and_Retrieval/

[2] GDC API file downloads

https://docs.gdc.cancer.gov/API/Users_Guide/Downloading_Files/

[3] NCI: citing TCGA

https://www.cancer.gov/ccg/research/genome-sequencing/tcga/using-tcga-data/citing

[4] Yang et al. EMT research consensus, 2020

https://www.nature.com/articles/s41580-020-0237-9

[5] Collins et al. TRIPOD+AI, BMJ 2024;385:e078378

https://www.bmj.com/content/385/bmj-2023-078378

Results are based on data generated by the TCGA Research Network: https://www.cancer.gov/tcga.
Real-data values: pilot_results.json and manifests. Clinical efficacy and biological validity remain unestablished.